

## Zepto E-Commerce SQL Data Analysis Project

### Project Overview

This project is a real-world SQL Data Analytics project based on an e-commerce inventory dataset from Zepto, one of India's fastest-growing quick-commerce platforms. The project demonstrates how SQL can be used for data exploration, cleaning, and business analysis in the retail and e-commerce domain.

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### Objectives

- Build and manage a relational database using PostgreSQL
  - Perform exploratory data analysis (EDA)
  - Clean and preprocess inventory data
  - Generate business insights using SQL queries
  - Simulate real-world e-commerce analytics workflows
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### Dataset Information

The dataset contains product inventory information scraped from Zepto's product listings. Each row represents a unique SKU (Stock Keeping Unit).

### Dataset Features

| Column Name            | Description               |
|------------------------|---------------------------|
| sku_id                 | Unique product identifier |
| category               | Product category          |
| name                   | Product name              |
| mrp                    | Maximum Retail Price      |
| discountPercent        | Discount percentage       |
| availableQuantity      | Stock quantity            |
| discountedSellingPrice | Final selling price       |

| Column Name | Description                 |
|-------------|-----------------------------|
| weightInGms | Product weight              |
| outOfStock  | Product availability status |
| quantity    | Package quantity            |

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### Technologies Used

- PostgreSQL
  - SQL
  - pgAdmin
  - CSV Dataset
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### Database Design

```
CREATE TABLE zepto (  
    sku_id SERIAL PRIMARY KEY,  
    category VARCHAR(120),  
    name VARCHAR(150) NOT NULL,  
    mrp NUMERIC(8,2),  
    discountPercent NUMERIC(5,2),  
    availableQuantity INTEGER,  
    discountedSellingPrice NUMERIC(8,2),  
    weightInGms INTEGER,  
    outOfStock BOOLEAN,  
    quantity INTEGER  
);
```

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### Project Workflow

## **1 Data Import**

- Imported CSV dataset into PostgreSQL
- Resolved UTF-8 encoding issues
- Verified successful data loading

## **2 Exploratory Data Analysis (EDA)**

- Checked total records and null values
- Identified unique product categories
- Compared in-stock vs out-of-stock products
- Detected duplicate product entries

## **3 Data Cleaning**

- Removed invalid pricing records
- Converted prices from paise to rupees
- Standardized numerical values

## **4 Business Analysis**

Generated insights using SQL queries:

- Top discounted products
- High-value out-of-stock products
- Revenue estimation by category
- Top categories with highest discounts
- Price-per-gram analysis
- Inventory weight analysis

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## **Challenges Faced**

- Handling UTF-8 encoding issues during CSV import
- Cleaning inconsistent pricing data
- Removing invalid records with zero values

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### Skills Demonstrated

- SQL Query Writing
- Data Cleaning
- Exploratory Data Analysis
- Business Analytics
- PostgreSQL Database Management

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### Conclusion

This project demonstrates practical SQL skills used in real-world e-commerce analytics. It highlights the process of transforming raw inventory data into meaningful business insights through data cleaning, exploration, and analytical querying.